

PHY 211

Lecture 10 Notes

By

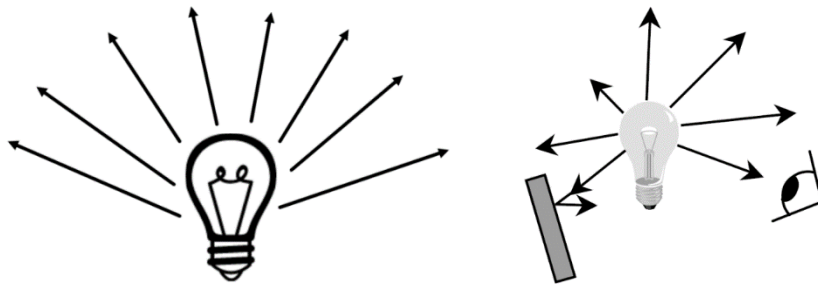
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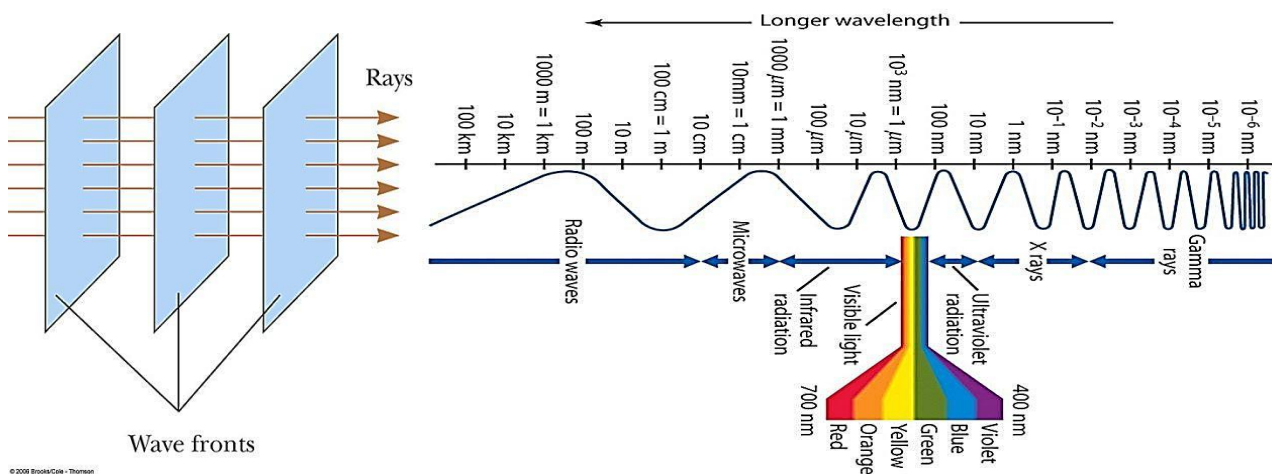
Chapter 9

The Nature of Light and the Principles of Ray Optics

- **Optics:** a Physics branch which studies light, its nature, properties and optical instruments.
- **Geometric Optics:** an optics branch which studies reflection and refraction of light.
- To describe reflection and refraction, we use the concept of light ray in which the light is represented by rays emitted from the source of light.



- **Ray:** a straight line drawn along the direction of propagation of the wave, it shows the path of the waves as it travels through space.



- Light can be considered as a stream of particles called photons, also light can be considered as electromagnetic waves (transverse waves, part of EM spectrum).
- So, light has **dual nature** that is in some cases it behaves like a wave and in some other cases it behaves like a particle.

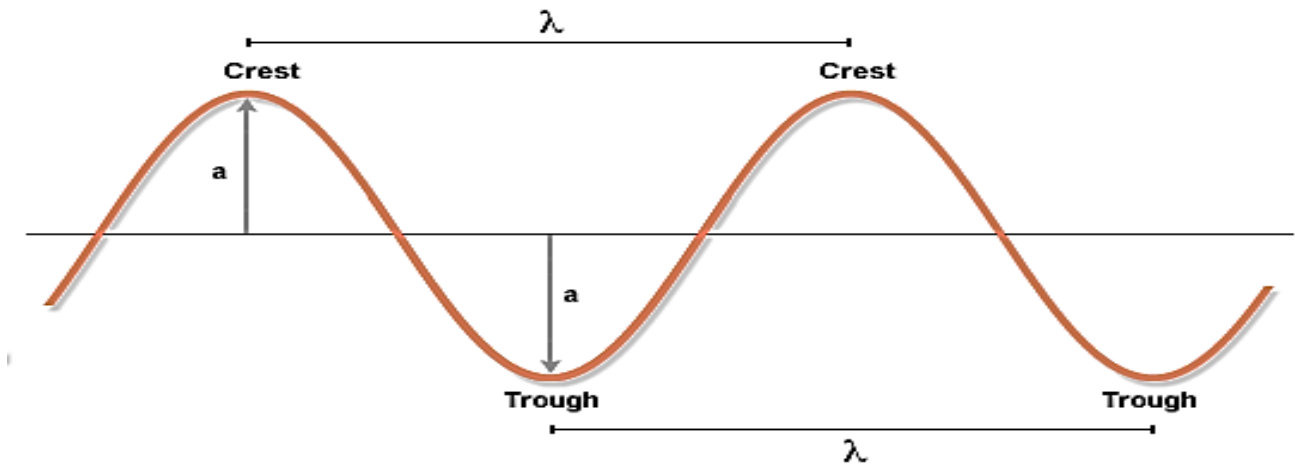
- For any wave:

$$v = \lambda f$$

v : speed of wave

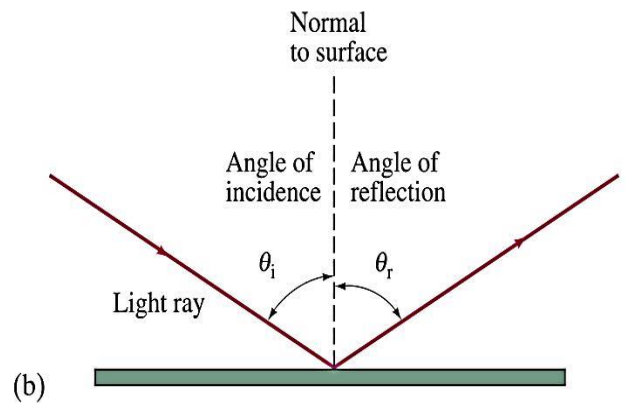
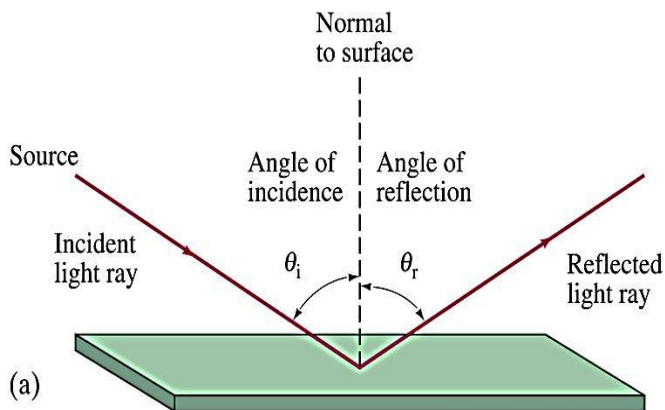
λ : wavelength (distance between two successive crests or troughs)

f : frequency (number of complete cycles in one sec)



➤ Reflection of light:

- It is the rebounding back of light when it meets a reflecting surface.

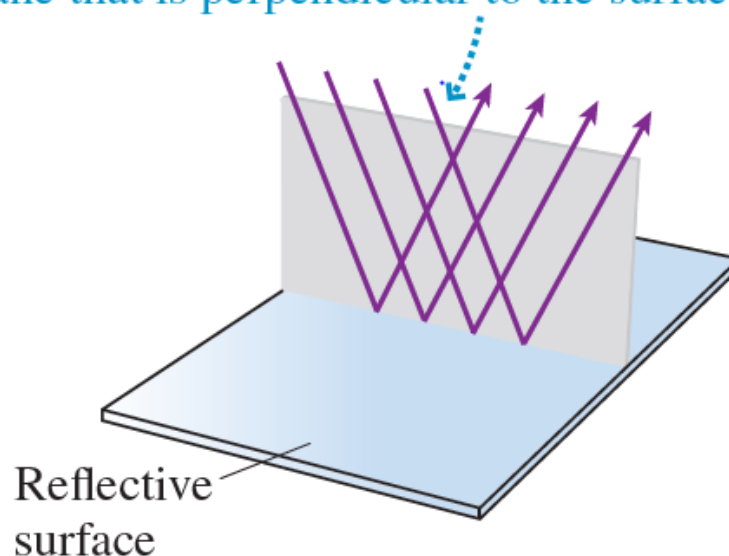


- Angle of incidence: angle between incident ray and normal to surface.
- Angle of reflection: angle between reflected ray and normal to surface
- Reflection rules (laws):

Rule I: *Angle of incidence (θ_i) = (θ_r) Angle of reflection*

Rule II: *The incident light ray, reflected light ray normal to reflecting surface must all lie in the same plane*

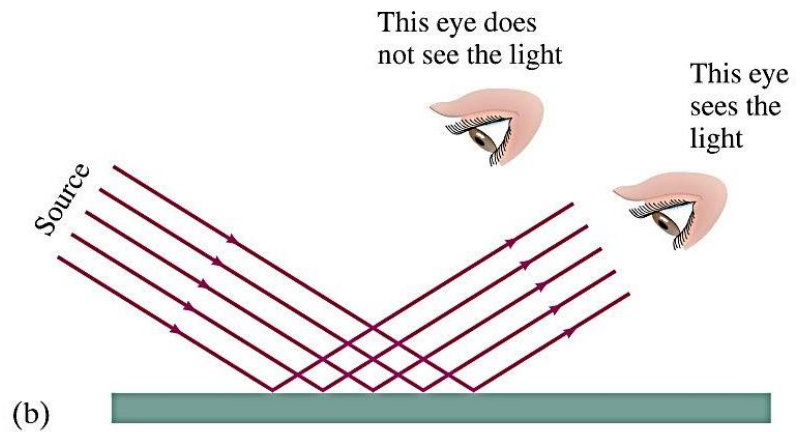
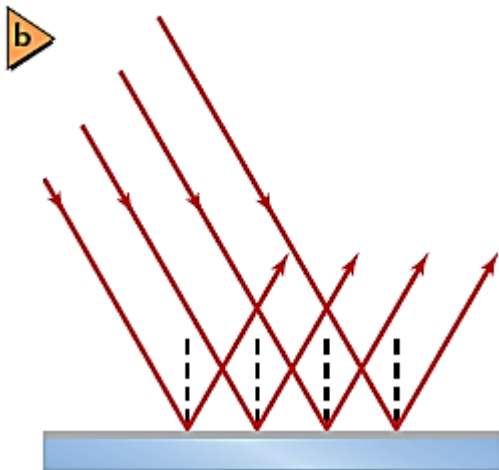
Both the incident and reflected rays lie in a plane that is perpendicular to the surface.



- **Types of reflection:**

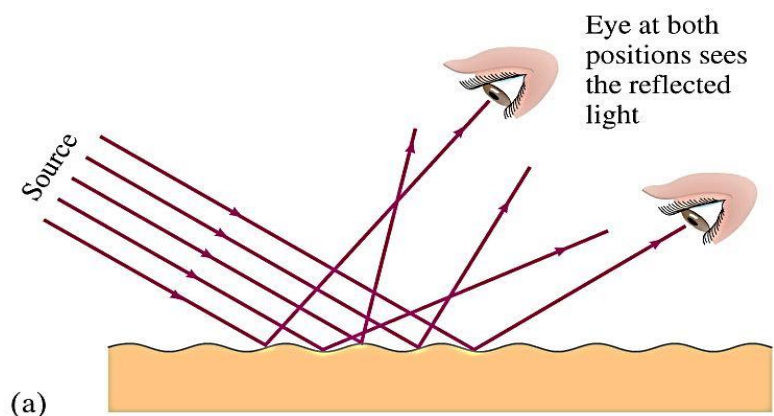
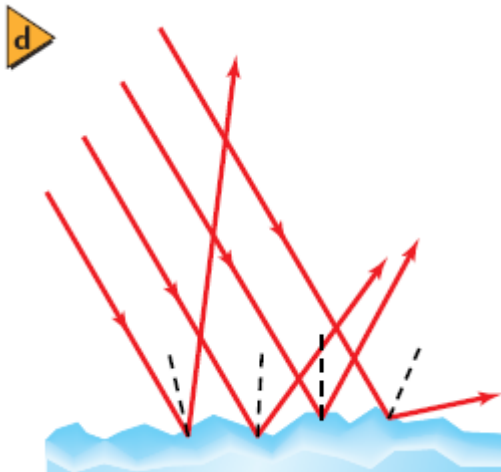
- 1- Specular reflection**

- Reflection on smooth surface.



- 2- Diffuse reflection**

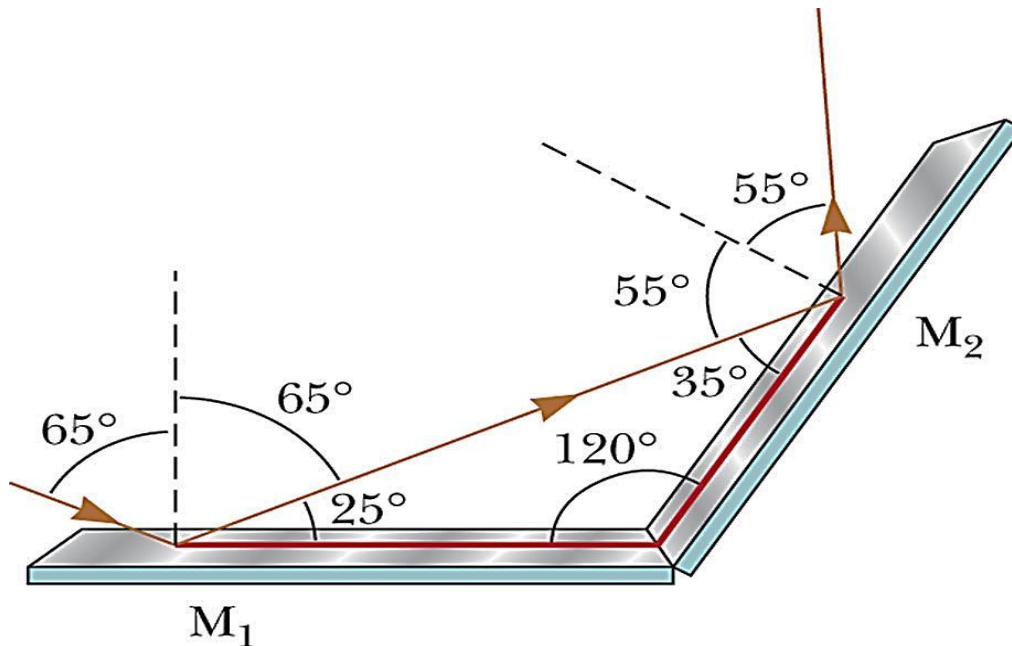
- Reflection on rough surface.
- When light reflects from a rough surface, the law of reflection still holds, but the angle of incidence varies. This is called diffuse reflection.



- **Example 1:**

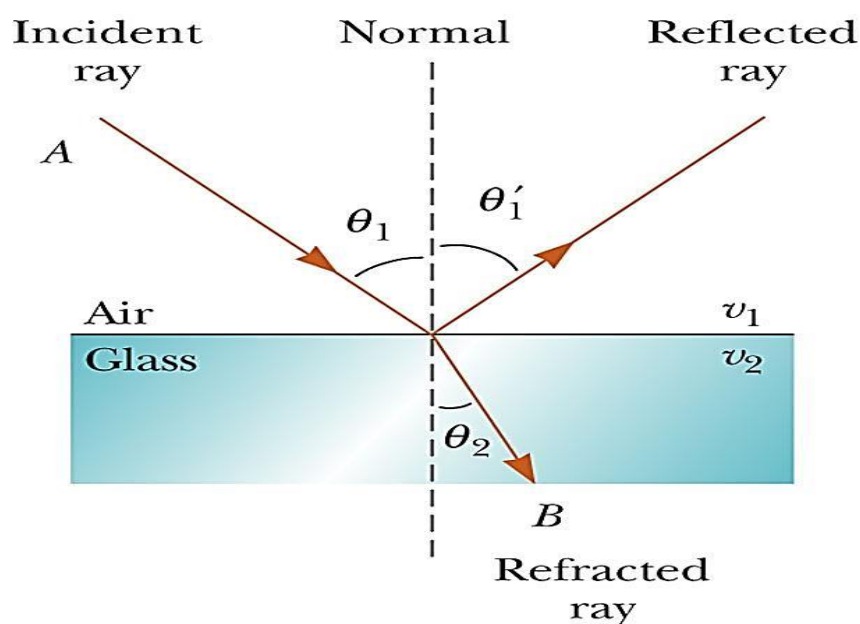
Two mirrors make an angle of 120° with each other. A ray is incident on mirror M_1 at an angle of 65° to the normal. Find the angle the ray makes with the normal to M_2 after it is reflected from both mirrors.

Solution



➤ **Refraction of light:**

- It is the change of direction of light as it passes from one medium to another.



- **Angle of refraction:** angle between refracted ray and normal to surface.

- **Refraction law:**

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

- The speed of light is higher in the less dense medium.
- **Angle of deviation:** Angle between refracted ray and extension of incident ray

$$\beta = \theta_1 - \theta_2$$

- **Refractive index (index of refraction) (n):** It is the ratio of the speed of light in air (vacuum) ($c = 3 \times 10^8 \frac{\text{m}}{\text{sec}}$) to the speed of light in medium.

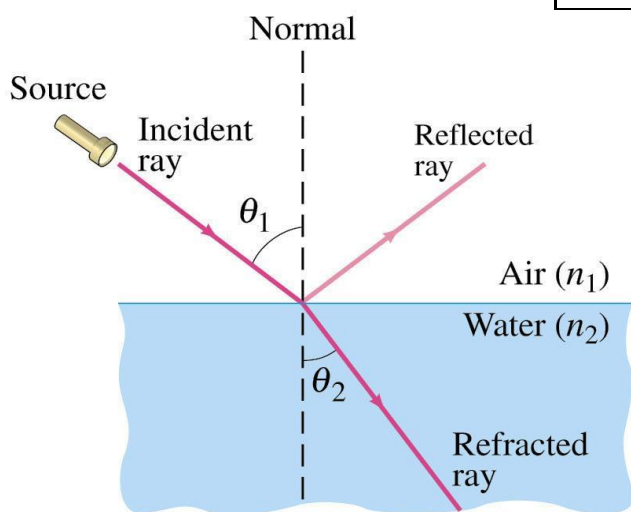
$$n = \frac{c}{v} \quad \text{unitless} \quad n \propto \frac{1}{v}$$

$$n = 1 \text{ for air} \quad \& \quad n > 1 \text{ for any other medium}$$

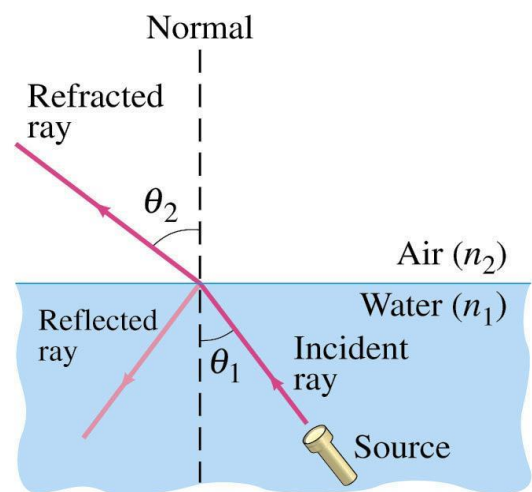
$$n_w = \frac{c}{v_w} \quad n_g = \frac{c}{v_g} \quad n_{oil} = \frac{c}{v_{oil}}$$

- **Relative refractive index between two mediums:** the ratio between the refractive indices of the two mediums

$${}_1n_2 = \frac{n_2}{n_1}$$



(a) $n_2 > n_1$
Ray bends toward $\perp$
 $v_2 < v_1$

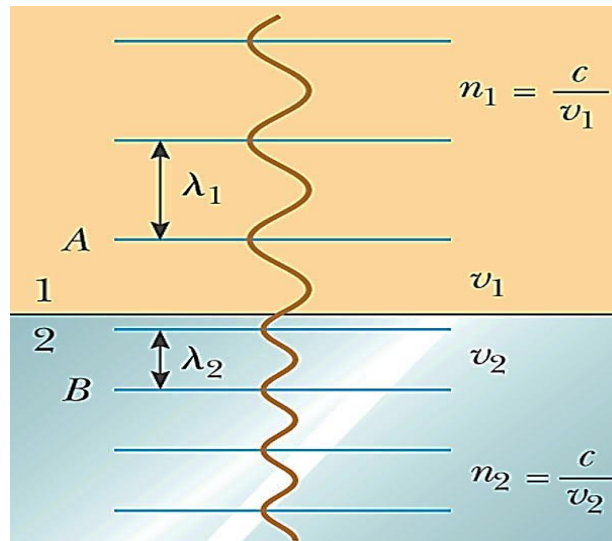


(b) $n_1 > n_2$
Ray bends away from $\perp$
 $v_2 > v_1$

- **Snell's law:**

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

- When the light passes from one medium to another, the speed v and the wavelength λ change while the **frequency f does not change**.



$$v_1 = \lambda_1 f \quad \& \quad v_2 = \lambda_2 f$$

$$\frac{n_1}{n_2} = \frac{v_2}{v_1} = \frac{\lambda_2}{\lambda_1}$$

$$n \propto \frac{1}{\lambda}$$

$$\lambda_m = \frac{\lambda_{air}}{n_m}$$

• Example 2:

A beam of light of wavelength 550 nm traveling in air is incident on a slab of transparent material, the incident beam makes an angle of 40° with the normal, and the refracted beam makes an angle of 26° with the normal. Find the index of refraction of the material.

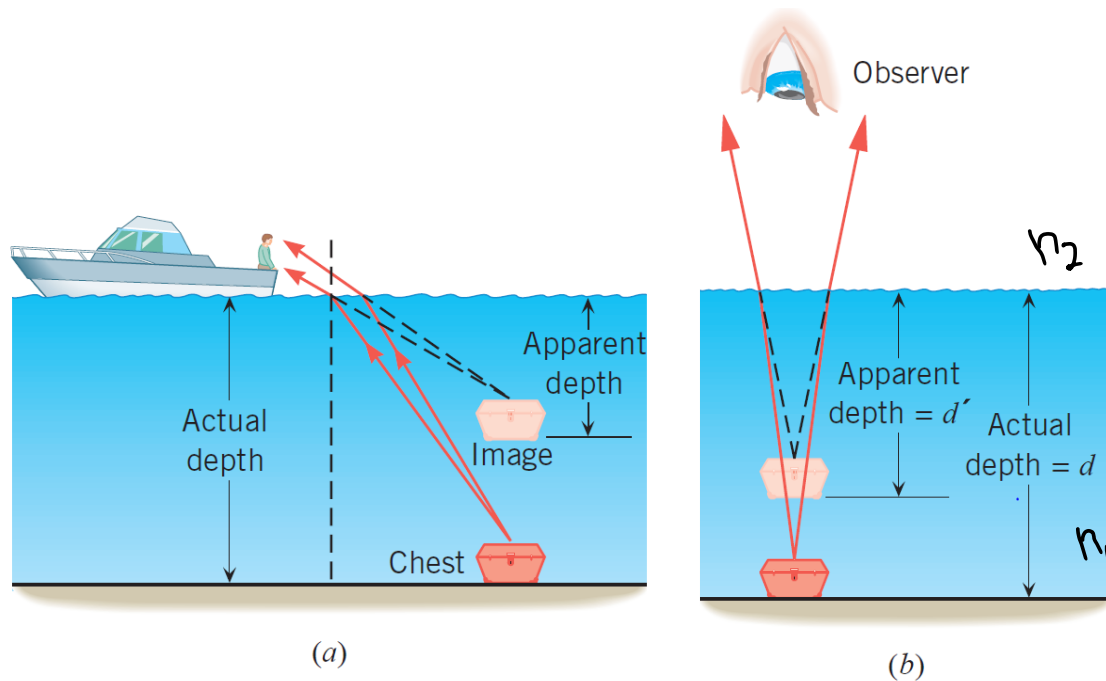
Solution

$$n_1 = 1 \quad \theta_1 = 40^\circ \quad \theta_2 = 26^\circ$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_2 = \frac{n_1 \sin \theta_1}{\sin \theta_2} = \frac{1 \times \sin 40}{\sin 26} = 1.47$$

• Apparent depth:



- An object lying under water appears to be closer to the surface than it actually is.
- For the situation shown in Figure “a”, it is difficult to determine the apparent depth.
- A much simpler case is shown in part “b”, where the observer is directly above the submerged object, and the apparent depth d' is related to the actual depth d by:

$$d' = d \frac{n_2}{n_1}$$

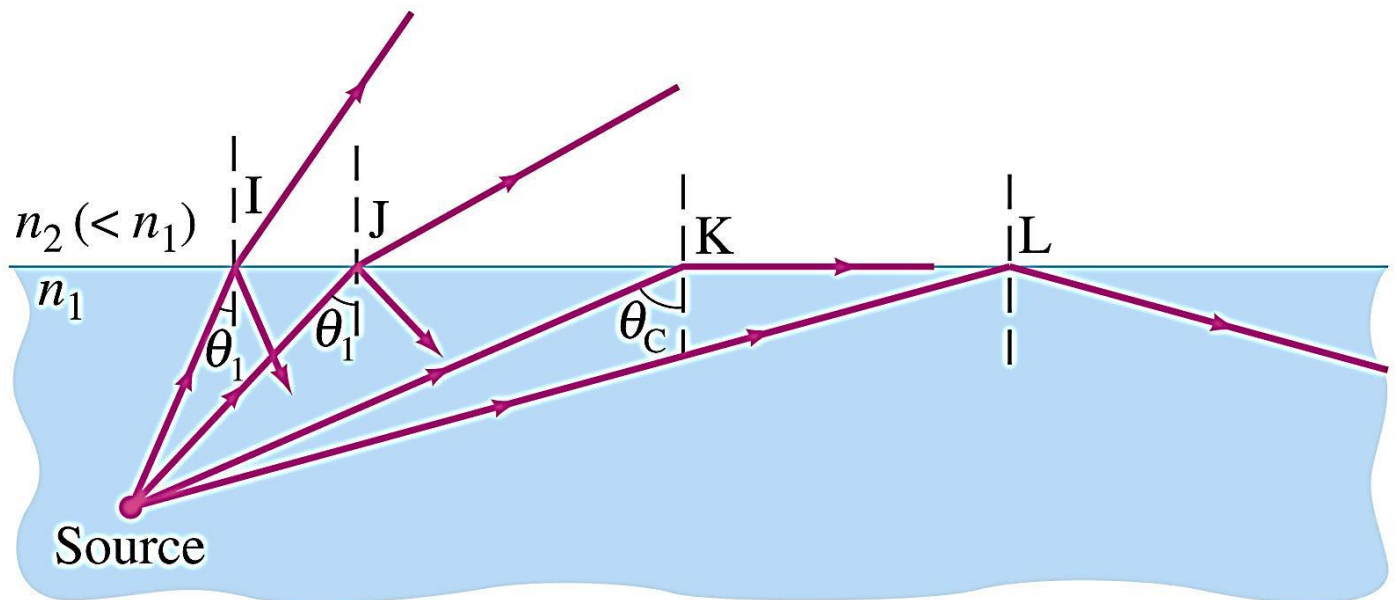
• **Example 3:**

A swimmer is treading water (with her head above the water) at the surface of a pool 3.00 m deep. She sees a coin on the bottom directly below. How deep does the coin appear to be?

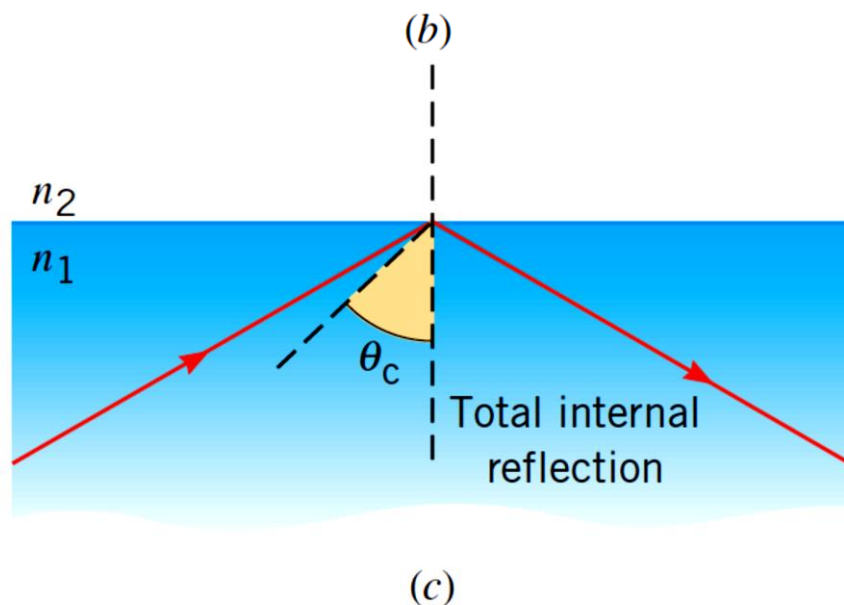
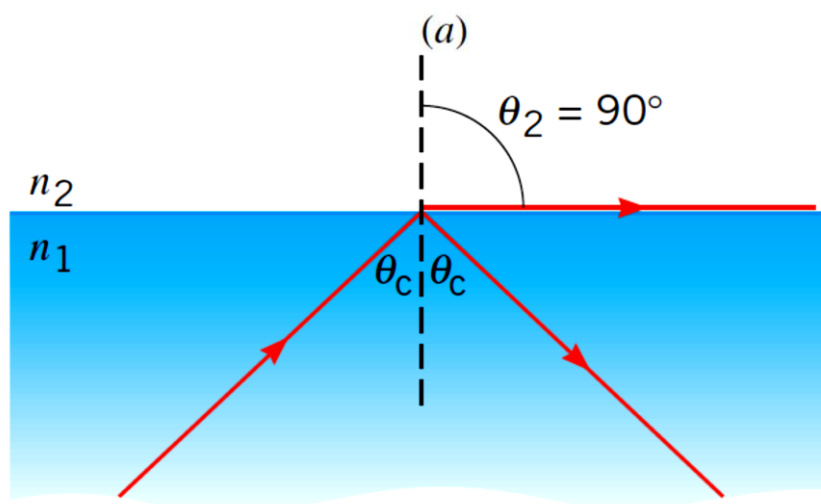
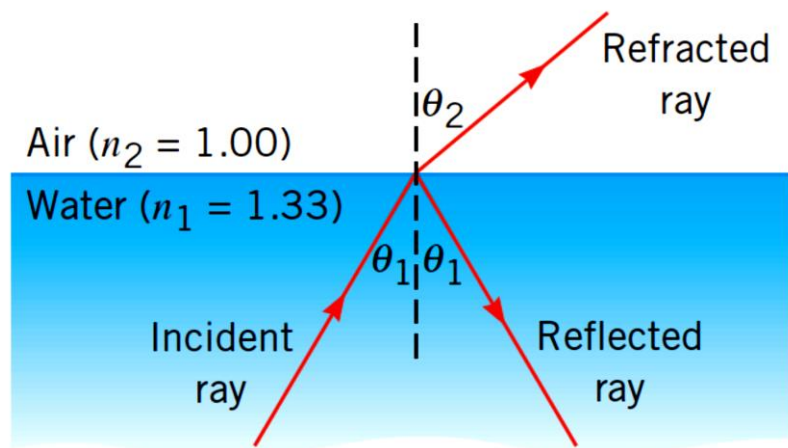
Solution

$$d' = d \frac{n_2}{n_1} = 3 \times \frac{1}{1.33} = 2.26 \text{ m}$$

➤ **Total internal reflection of light:**



- When light passes from a medium of larger refractive index into one of smaller refractive index (for example, from water to air) the refracted ray bends *away* from the normal, as in Figure a. As the angle of incidence increases, the angle of refraction also increases.
- When the angle of incidence reaches a certain value, called the **critical angle** θ_c , the angle of refraction is 90° . Then the refracted ray points along the surface; Figure b.
- When the angle of incidence exceeds the critical angle, as in Figure c, there is no refracted light. All the incident light is reflected back into the medium from which it came, a phenomenon called **total internal reflection**.
- Total internal reflection occurs only when light travels from a higher index medium toward a lower-index medium. It does not occur when light propagates in the reverse direction—for example, from air to water.
- **Critical Angle θ_c :** It is the angle of incidence in the medium of higher refractive index corresponding to angle of refraction of 90° in the medium of lower refractive index.



- Applying Snell's law:

$$n_1 \sin \theta_c = n_2 \sin 90$$

$$\sin \theta_c = \frac{n_2}{n_1} \longrightarrow \theta_c = \sin^{-1} \frac{n_2}{n_1}$$

- **Example 4:**

A beam of light is propagating through diamond ($n_1 = 2.42$) and strikes a diamond–air interface at an angle of incidence of 28° .

- (a) Will part of the beam enter the air ($n_2 = 1$) or will the beam be totally reflected at the interface?
- (b) Repeat part (a), assuming that the diamond is surrounded by water ($n_2 = 1.33$) instead of air.

Solution

(a)

$$\theta_c = \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} \frac{1}{2.42} = 24.4^\circ$$

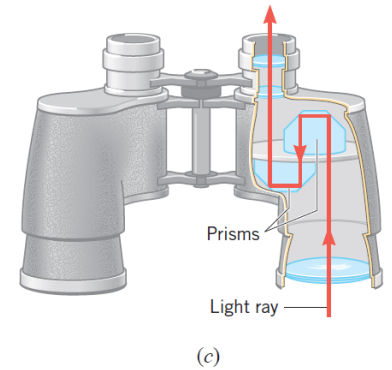
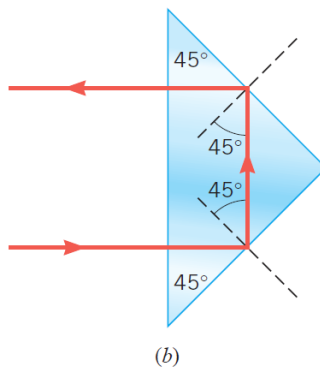
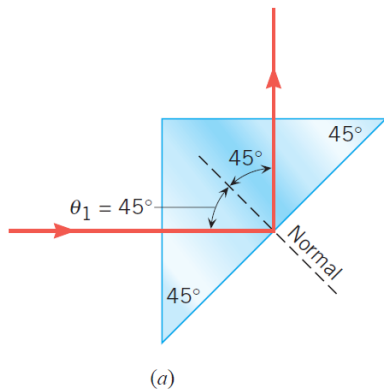
Because the angle of incidence of 28° is greater than the critical angle, there is no refraction, and the light is totally reflected back into the diamond.

(b)

$$\theta_c = \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} \frac{1.33}{2.42} = 33.3^\circ$$

Now a beam of light that has an angle of incidence of 28° (less than the critical angle of 33.3°) at the diamond–water interface is refracted into the water. (partially refracted - partially reflected)

- Applications on total internal reflection:**



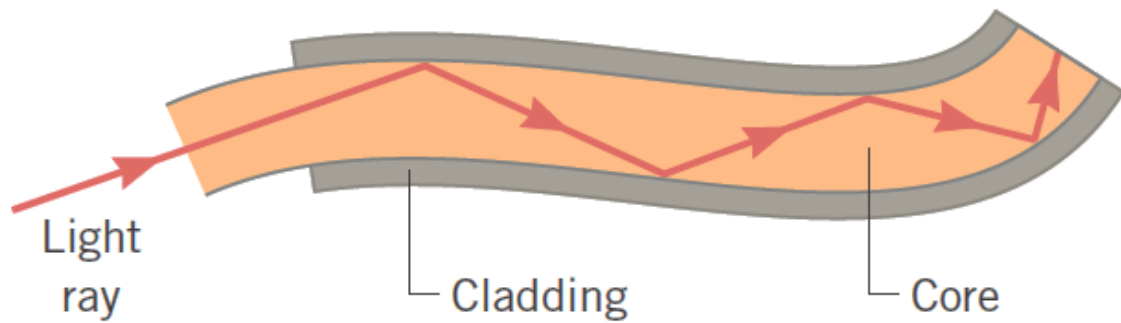
- Many optical instruments, such as binoculars, periscopes, and telescopes, use glass prisms and total internal reflection to turn a beam of light through 90° or 180°. Figure (a) shows a light ray entering a 45° – 90° – 45° glass prism ($n_1 = 1.5$) and striking the hypotenuse of the prism at an angle of incidence of $\theta_1 = 45^\circ$.

- The critical angle for a glass–air interface is

$$\theta_c = \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} \frac{1}{1.5} = 42^\circ$$

- Since the angle of incidence is greater than the critical angle, the light is totally reflected at the hypotenuse and is directed vertically upward in the drawing, having been turned through an angle of 90°.
- Part (b) of the picture shows how the same prism can turn the beam through 180° when total internal reflection occurs twice.
- Figure (c) illustrates such an application in binoculars.
- An important application of total internal reflection occurs in fiber optics, where hair-thin threads of glass or plastic, called optical fibers, “pipe” light from one place to another.
- The figure shows that an optical fiber consists of a cylindrical inner core that carries the light and an outer concentric shell, the cladding. The core is made from transparent glass or plastic that has a relatively high index of refraction. The cladding is also made of glass, but of a type that has a relatively low index of refraction.

- Light enters one end of the core, strikes the core/cladding interface at an angle of incidence greater than the critical angle, and, therefore, is reflected back into the core. Light thus travels inside the optical fiber along a zigzag path.



- In the field of medicine, optical fiber cables have had extraordinary impact. In the practice of endoscopy, for instance, a device called an endoscope is used to peer inside the body.